

batteries, it's essential for both you and me to grasp the critical role that electrolyte impurities play in hampering their performance. These impurities act as barriers, obstructing the movement of lithium ions across the separator, which significantly undermines their ability to efficiently integrate into the electrodes.

What does this mean for us? We witness a marked slowdown in the battery's charging and discharging rates, leading to a diminished lifespan. Moreover, these impurities can trigger side reactions that expedite the degradation of both electrolytes and electrodes, resulting in longer charging times and a reduced range for electric vehicles (EVs).

The pattern of batteries' degradation varies greatly, depending on whether they employ liquid, semi-solid, or solid electrolytes. The market offers a plethora of electrolyte formulations, each presenting its own set of pros and cons for different battery technologies. A primary focus for research and development (R&D) teams, including ours, is pinpointing which specific components are responsible for this accelerated degradation. Identifying these components, which can be efficiently monitored through UV/Vis spectrophotometry, is pivotal in the quest for safer and more enduring batteries.

As batteries endure multiple cycles, their electrolytes deteriorate due to a variety of chemical and electrochemical reactions. This degradation manifests through changes in the absorption spectrum. By consistently observing these spectral shifts, we can track the progression of battery degradation and anticipate its impact on crucial performance indicators, such as battery capacity.

Yet, establishing a reliable reference curve poses a significant

Streamlining lab analysis

Integrating multi-parameter analyses, like titration and pH measurement, into a single analysis workflow can significantly enhance laboratory efficiency and accuracy. This approach offers several key benefits, particularly when utilising advanced analytical systems.

- **Reduced sample volume.** Integration cuts down sample volume for techniques like titration and pH measurement, reducing waste and conserving resources, crucial for limited or precious samples.
- **Time savings.** Independent systems boost time efficiency by reducing manual work, freeing operators for other tasks and speeding up analysis cycles.
- **Streamlined data management.** Multi-parameter analysis integration simplifies data management by merging results into one report, improving clarity, reducing errors, and easily integrating with quality systems.

The integration of multi-parameter analyses into a single workflow offers significant benefits in terms of efficiency, resource management, and data handling. By adopting such advanced systems, laboratories can optimise their operations, ensure high-quality results, and maintain compliance with stringent quality standards.

challenge. Electrolytes are delicate, and factors like manufacturing processes and transportation can compromise their quality. Even mere exposure to air can accelerate electrolyte degradation. To counter these variables, adopting a strategy that employs multiple representative samples is wise. This method enables the creation of a statistical average spectrum, providing a more precise and comprehensive insight into the electrolyte's behaviour over time and under various conditions.

Colour changes in battery electrolytes

When dealing with electrolyte degradation, it is often signalled by a colour change, typically going from clear to shades of yellow or brown. To quantify these changes, we rely on several standardised colour scales like APHA, Pt/Co, Hazen, or the yellowness-index. These scales convert the colour variation into a single number, making it easier to assess the electrolyte's quality.

In the past, determining the colour of electrolytes was something we had to do by eye, which was tricky due to variations in lighting and personal perception. Now, to ensure consistent results that are not dependent on the

individual conducting the test, many standards employ UV/Vis spectrophotometry. This method is standardised and thoroughly tested to ensure accuracy, largely depending on the spectrophotometer's performance.

UV/Vis spectrophotometry has become a handy tool, especially during the development phase of electrolyte solutions. When needing to detect very small impurities, we might use other methods like ICP-OES. For routine measurements, basic UV/Vis spectrophotometers that are easy to use are our go-to. And if in a lab looking to boost efficiency and automation, we consider advanced models. Models like the UV5 from METTLER TOLEDO come with features like flow cells, cuvette changers, or autosamplers, enabling us to automate processing up to 303 samples in a single run.

Quick quality checks with advanced spectrophotometers

Modern UV/Vis spectrophotometers are all about making our lives easier. They've got these built-in colour scales which really cut down on the manual work and calculations we must do. This not only makes the instruments super user-friendly for you and me but also means we are